L10 First Order Applications

Sections 2.4, 3.1-3.3

Orthogonal trajectories

A geometric problem of interest in engineering is that of finding a family of curves (orthogonal trajectories) that intersects a given family of curves orthogonally at each point. Some applications are as follows:

perpendicular

electromagnetic theory–given lines of force of electric field, find the equipotential flux

heat conduction–given lines of constant temperature (isotherms), find lines of heat flow

fluid mechanics–given fluid flow streamlines, find the associated equipotential lines along which the fluid potential is constant

The slope of a curve orthogonal (perpendicular) to a given curve is the negative reciprocal of the original curve's slope.

$F(x,y)=k$

Curves orthogonal to the family $F(x,y)=k$ satisfy the DE

$$\frac{\partial F}{\partial y}dx - \frac{\partial F}{\partial x}dy = 0$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

orthogonal

new slope

$$\frac{dy}{dx} = \frac{F_y}{F_x}$$

$$\Rightarrow F_y dy = -F_x dx$$

$$\Rightarrow F_x dx + F_y dy = 0$$

$$\Rightarrow F_x dy = F_y dx$$

$$\Rightarrow F_y dx - F_x dy = 0$$

ex. Find orthogonal trajectories to the family of hyperbolas $xy = k$.

$F(x,y) = xy$, $F_y = x$, $F_x = y$ $\Rightarrow y = \frac{k}{x}$

So $F_y dx - F_x dy = 0 \Rightarrow x\,dx - y\,dy = 0$

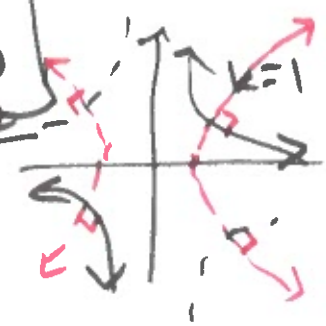
exact $M_y = 0$ $N_x = 0$

separable

$\Rightarrow \int x\,dx = \int y\,dy = \frac{1}{2}x^2 + C = \frac{1}{2}y^2$

$\Rightarrow y^2 - x^2 = C$

hyperbolas



Exponential Growth and Decay

Suppose the rate of change of a population of organisms is proportional to that quantity:

$$\frac{dP}{dt} = kP$$

autonomous (seperable)

This is a separable equation. We get

$$\Rightarrow \int \frac{1}{P}\,dP = \int k\,dt$$

$$\Rightarrow \ln|P| = kt + C$$

$$P = Ce^{kt}, \quad C = P_0 \leftarrow (P(0) = P_0)$$

We have exponential growth when $k > 0$ and exponential decay when $k < 0$.

Ex: Given an electric field $\vec{E} = \langle E_x, E_y \rangle$, its potential at a point $\vec{r}$ is given by

$$V = -\int_C \vec{E} \cdot d\vec{r} = -\int_C E_x\,dx + E_y\,dy$$

$$= \int_C V_x\,dx + V_y\,dy$$

we have

$$V_x = -E_x \ , \ V_y = -E_y$$

If C is closed then

Green's Theorem

$$0 = \oint_C \overset{P}{E_x}\,dx + \overset{Q}{E_y}\,dy = \iint_D (E_y)_x - (E_x)_y \, dA$$

$$\Rightarrow 0 = -\iint_D (V_{yx} - V_{xy})\,dA \Rightarrow V_{yx} - V_{xy} = 0$$

$$\Rightarrow V_{xy} = V_{xy}$$

So $E_x dx + E_y dy = 0$ is exact! Its solutions are called the <u>equipotential</u> lines of the electric field.

<u>Ex</u>: If we have a charged wire with charge Q

then the electric field is

$$\langle \underbrace{\frac{kQx}{x^2+y^2}}_{E_x}, \underbrace{\frac{kQy}{x^2+y^2}}_{E_y} \rangle$$

$$E_x dx + E_y dy = 0 \Rightarrow kQ\left(\frac{x}{x^2+y^2}dx + \frac{y}{x^2+y^2}dy\right) = 0$$

is exact $\Rightarrow V(x,y) = x^2 + y^2 = C$

The <u>electric field lines</u> are the orthogonal trajectories of the equipotential lines. In this case

$$kQ\left(\frac{y}{x^2+y^2}dx - \frac{x}{x^2+y^2}dy\right) = 0 \Rightarrow y = Cx$$

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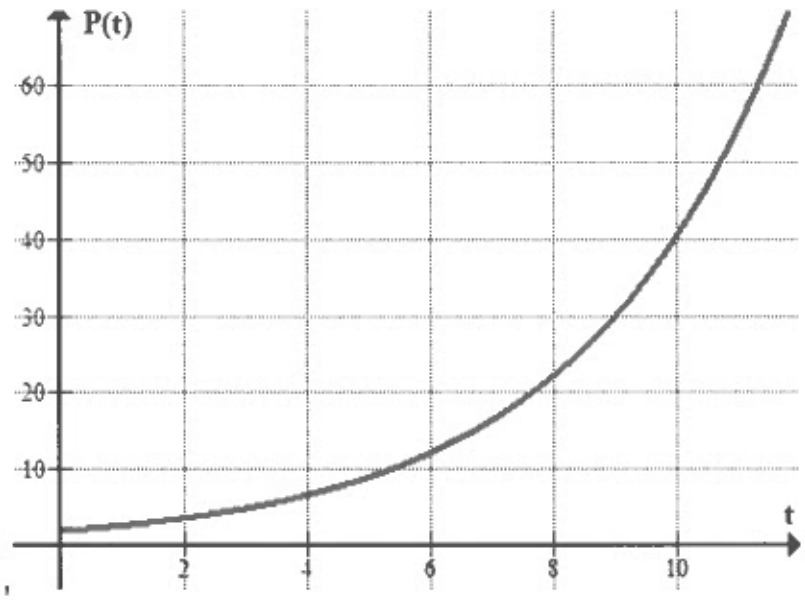
fluid flow $\langle u, v \rangle$ streamlines $v\,dx - u\,dy = 0$

growth.



Applications:

bank interest

radioactive decay

detecting art forgeries

population growth (people, bacteria, rabbits, etc.)

ex. The half-life of radium-226 is approximately 1590 years. A sample has a mass of 100 g of radium-226. Find

1. a formula for the mass of radium-226 remaining after t years
2. the mass after 1000 years
3. the time when the mass will be reduced to 30 g.

1. $m = m_0 e^{kt} = 100 e^{kt}$ if $m(1590) = 50$ then

$$50 = 100 e^{1590k} \Rightarrow e^{1590k} = \frac{1}{2} \Rightarrow k = \frac{1}{1590} \ln\left(\frac{1}{2}\right)$$

so $$m = 100 e^{\frac{1}{1590} \ln\left(\frac{1}{2}\right) t} = 100 \left(\frac{1}{2}\right)^{\frac{t}{1590}}$$

2. $$m(1000) = 100 \left(\frac{1}{2}\right)^{\frac{1000}{1590}} \approx 100 \left(\frac{1}{2}\right)^{0.6289} \approx 64{,}667 \text{ g}$$

3. $$30 = 100 \left(\frac{1}{2}\right)^{\frac{t}{1590}} \Rightarrow \left(\frac{1}{2}\right)^{t/1590} = \frac{3}{10}$$

$$\Rightarrow \frac{t}{1590} = \frac{\ln(3/10)}{\ln(1/2)}$$

$$\Rightarrow t = 1590 \left(\frac{\ln(3/10)}{\ln(1/2)}\right) \approx 2762 \text{ years}$$

$P = P_0 e^{kt}$ $\quad P_0 = 1000 \quad t=0$

$\Rightarrow 2000 = 1000 e^{k(1)}$ $\quad P = 2000 \quad t=1$

ex. You start with 1000 bacteria. The number of bacteria double every twenty minutes. How many bacteria do you have in 48 hours?

Let t represent each 20 minute period. 48 hours is 144 20-minute periods

$\Rightarrow 2 = e^{k} \Rightarrow k = \ln(2)$

$N(t) = 1000 e^{\ln(2)t} = 1000 e^{\ln(2^t)} = 1000 \cdot 2^t$

$N(144) = 1000 \cdot 2^{144} \approx 2.23 \times 10^{46}$

Weight of one bacterium (E. coli) $\approx$ 10^{-15} kg

total mass of bacteria: 2.23×10^{31} kg

weight of planet earth 5.972×10^{24} kg

If your population were bacteria, after 48 hours of exponential growth, their weight would exceed the weight of the entire planet. Obviously this model is not realistic. In real life there are limits to food, adverse effects from the accumulation of waste products, deaths, etc. More realistic population growth models take such limitations into account.

Logistic Models

Verhulst (1936) proposed the logistic model to account for limits to growth in living systems. His logistic differential equation has two positive parameters: r, a growth parameter, and K, the carrying capacity. His equation is

as $P \to K$ $\quad \frac{dP}{dt} \to 0$

$$\frac{dP}{dt} = rP(1 - \frac{P}{K})$$

autonomous

This is a separable equation:

$$\frac{dP}{P(1-\frac{P}{K})} = rdt$$

which has solution

$$\boxed{P(t) = \frac{P_0 K}{P_0 + (K - P_0)e^{-rt}}}$$

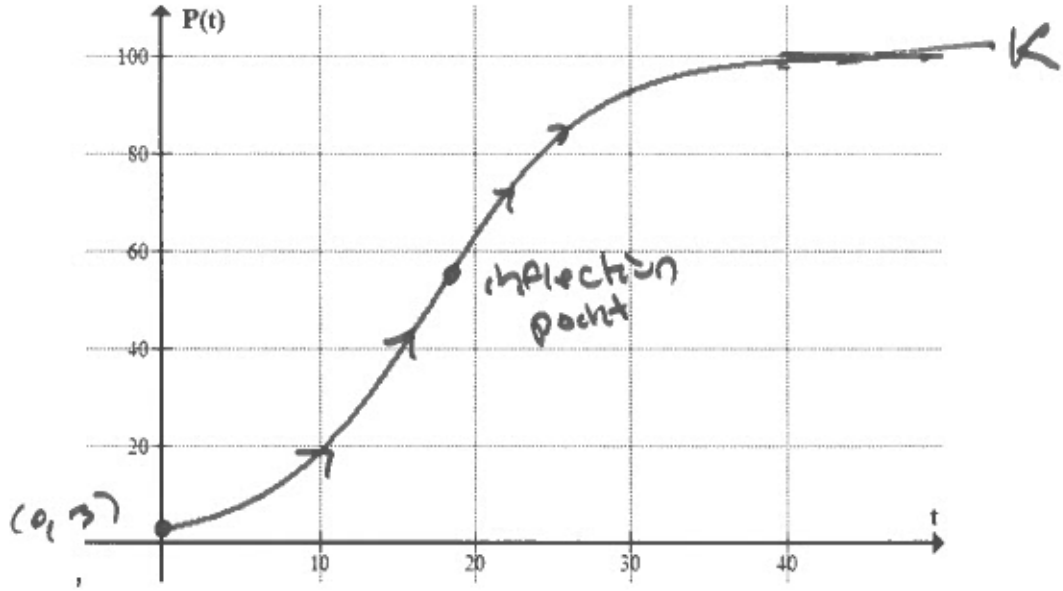
where P_0 is the initial population.

P_0

ex. Suppose an initial population of bacteria is 3 mg, the carrying capacity is 100 mg, and the growth parameter is 0.2/hr.

K $\quad r$

$$P(t) = \frac{3 \cdot 100}{3 + (100 - 3)e^{-0.2t}} = \frac{300}{3 + 97e^{-0.2t}}$$



ex. In the logistic model for population growth, $\frac{dP}{dt} = P(6-2P)$. What is the carrying capacity of the population?

$$\frac{dP}{dt} = P(6-2P) = 6P\left(1-\frac{2P}{6}\right) = \underset{r}{6}P\left(1-\frac{P}{\underset{K}{3}}\right)$$

carrying capacity is 3.

Newton's Law of Cooling

If $\frac{dT}{dt} = -kT$ decay

Newton's Law of Cooling states that

$\frac{dT}{dt} = k(T_e - T)$ where k is a positive constant. Here,

altered autonomous

T is the temperature of an object at time t

T_0 is the initial temperature of an object at time $t = 0$

T_e is the surrounding (ambient) temperature

k is a constant governing the rate of cooling (usually we solve for it).

This is a separable equation.

$$\frac{dT}{T_e - T} = kdt$$

$$-\ln|T_e - T| = kt + C$$

$$e^{\ln|T_e - T|} = e^{-kt - C}$$

$$T - T_e = Ce^{-kt}$$

$T(0) = T_0$

Applying initial conditions we get

$$T(t) = T_e + (T_0 - T_e)e^{-kt}$$

Ex: Suppose you brew a cup of coffee. When made the coffee is at $90°C$ and the temperature in the room is around $20°C$. After 5 minutes the temperature of the coffee is $80°C$. How long until the coffee is at $65°C$?

$$T(t) = T_e + (T_0 - T_e)e^{-kt}$$

$T_e = 20°C$, $T_0 = 90°C$ $\Rightarrow$ $T(t) = 20 + (90-20)e^{-kt}$
$= 20 + 70e^{-kt}$

$T(5) = 80$

$\Rightarrow$ $80 = 20 + 70e^{-5k}$

$\Rightarrow$ $60 = 70e^{-5k}$ $\Rightarrow$ $e^{-5k} = \frac{6}{7}$ $\Rightarrow$ $-5k = \ln(\frac{6}{7})$
$= k = -\frac{1}{5}\ln(\frac{6}{7})$
≈ 0.031

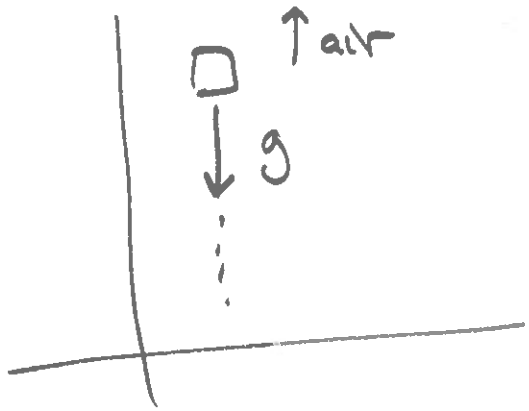
$$T(t) = 20 + 70e^{-0.031 \cdot t}$$

at $65° = 20 + 70e^{-0.031 \cdot t}$ $\Rightarrow$ $45 = 70e^{-0.031 \cdot t}$

$\Rightarrow$ $\frac{45}{70} = e^{-0.031 \cdot t}$

$\Rightarrow$ $t = -\frac{1}{0.031}\ln\left(\frac{45}{70}\right) \approx 14.25$ minutes.

Falling



$$F = ma = \underbrace{m \cdot \frac{dv}{dt}} = F_g = \underbrace{mg}$$

$$\frac{dv}{dt} = g$$

Equation

air resistance

$$m\frac{dv}{dt} = F_g - F_a = mg - kv$$

$$\Rightarrow \quad m\frac{dv}{dt} = mg - kv \qquad \underline{\text{linear}}$$

$$\Rightarrow \quad \frac{dv}{dt} = g - \frac{k}{m}v \Rightarrow \frac{dv}{dt} + \frac{k}{m}v = g$$

$P(t)$ $\rightarrow Q$

$$\mu(t) = e^{\int \frac{k}{m}dt} = e^{\frac{k}{m}t}$$

$$\Rightarrow \quad v = \frac{1}{\mu}\left[\int \mu Q \, dt + C\right]$$

$$= e^{-\frac{k}{m}t}\left[\int e^{\frac{k}{m}t} \cdot g \, dt + C\right]$$

$$\Rightarrow \quad v = e^{-\frac{k}{m}t}\left[\frac{m}{k}e^{\frac{k}{m}t} \cdot g + C\right]$$

$$\Rightarrow \quad v = \frac{mg}{k} + Ce^{-\frac{k}{m}t}$$

$v(0) = 0$

$$\Rightarrow \quad C = -\frac{mg}{k}$$

$$\Rightarrow \quad \boxed{v = \frac{mg}{k}\left(1 - e^{-\frac{k}{m}t}\right)}$$

$t \to \infty$

$v \to \frac{mg}{k}$ terminal velocity